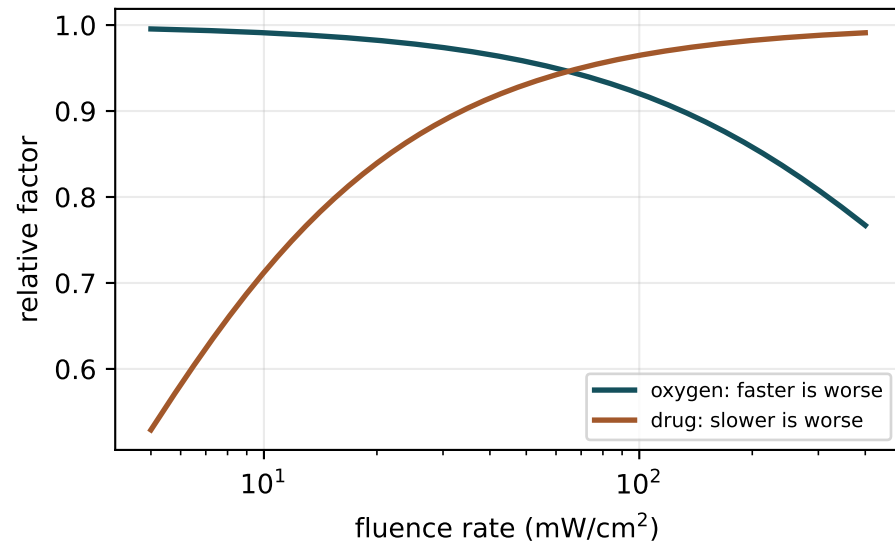
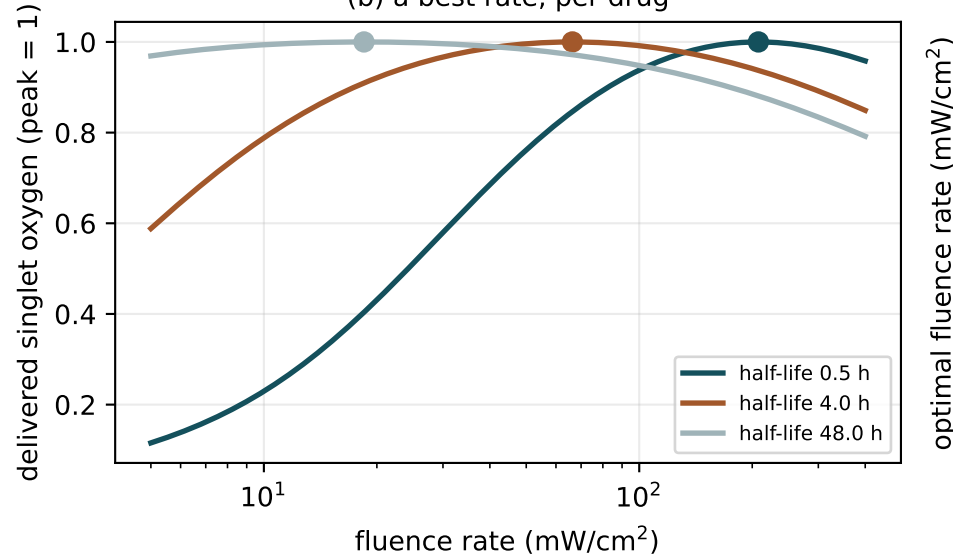


Photodynamic therapy consumes the oxygen it needs, so the same light delivered faster does less

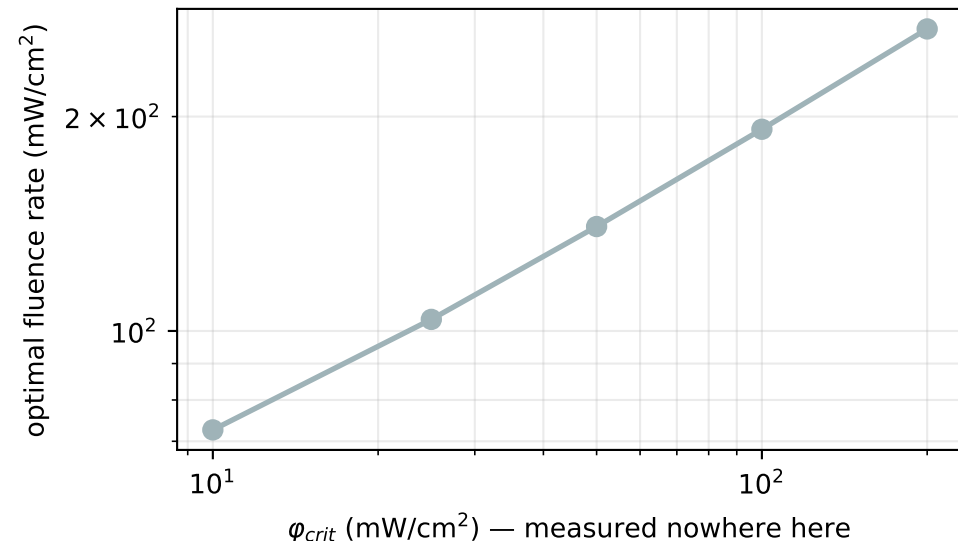
(a) two monotonic effects, opposed



(b) a best rate, per drug



(c) the optimum rides on it, exponent 0.43



A Type II photosensitizer works by handing energy to ground-state oxygen, so it depletes its own substrate at the site it is treating; Henderson & Busch (PMID 16615136) report depletion within seconds at 75 mW/cm². Against that, a slower illumination runs further into the sensitizer's own clearance, and drug that has left cannot be excited - which is not new machinery here but the module's existing pharmacokinetics, integrated across the illumination instead of sampled once at its start. Sampling once is exactly the approximation that makes a long treatment look free. (c) is the refusal: the optimum's POSITION scales as roughly the square root of a parameter nothing in this repository measures, so the direction is the result and the milliwatt figure is a restatement of an assumption.